Convexity of hypergraph polynomials from exponential minima

Principia Math In Progress

21 September 2026

Task 1: MathDB 347913. Let $E = \binom{[n]}{L}$, $m = |E|$, and

$$f_d(x) = \sum_{(e_1,\dots,e_d)\in E^d} \prod_{k=1}^{d} \frac{x_{e_k}}{|e_1 \cup \cdots \cup e_k|}. \tag{1}$$

Conjecture 1 of Brosch, Laurent and Steenkamp [1] asks whether f_d is convex on $\Delta_m = \{x \geq 0 : \sum_e x_e = 1\}$ for every $n, L, d \geq 2$ (with $L \leq n$, so the simplex is nonempty). We prove this, and in fact prove strict convexity on the whole nonnegative orthant for any family of distinct nonempty hyperedges.

Theorem 1. *Let V be a finite set and E a nonempty family of distinct nonempty subsets of V. Define f_d by (1), with $d \geq 2$. Let $(T_v)_{v\in V}$ be independent exponential random variables of rate one, and set $Y_e = \min_{v\in e} T_v$. Then*

$$\boxed{f_d(x) = \frac{1}{d!}\mathbb{E}\left[\left(\sum_{e\in E} x_e Y_e\right)^d\right]} \qquad (x \in \mathbb{R}^E). \tag{2}$$

Consequently f_d is strictly convex on $\mathbb{R}^E_+$. Its Hessian is positive definite at every nonzero $x \in \mathbb{R}^E_+$; for $d = 2$ it is positive definite at the origin as well. When $E = \binom{[n]}{L}$, the uniform vector is the unique minimizer on Δ_m.

Lemma 1. *For any ordered tuple $(e_1, \dots, e_d)$ of nonempty edges, including tuples with repeated edges,*

$$\mathbb{E}\prod_{i=1}^{d} Y_{e_i} = \sum_{\pi\in S_d} \prod_{k=1}^{d} \frac{1}{|e_{\pi(1)} \cup \cdots \cup e_{\pi(k)}|}. \tag{3}$$

Proof. By the tail-integral formula and Tonelli's theorem, the left side is

$$\int_{[0,\infty)^d} \mathbb{P}(Y_{e_i} > s_i \text{ for all } i)\, ds = \int_{[0,\infty)^d} \exp\left(-\sum_{v\in V} \max_{i:v\in e_i} s_i\right) ds,$$

where an empty maximum is zero. This probability formula follows because T_v must exceed every s_i for which $v \in e_i$. Partition the integration region, up to measure-zero ties, into the $d!$ chambers $s_{\pi(1)} \geq \cdots \geq s_{\pi(d)} \geq 0$. In one chamber put $u_k = s_{\pi(k)} - s_{\pi(k+1)}$ and $s_{\pi(d+1)} = 0$. The Jacobian is one, and

$$\sum_{v\in V} \max_{i:v\in e_i} s_i = \sum_{k=1}^{d} |e_{\pi(1)} \cup \cdots \cup e_{\pi(k)}|\, u_k.$$

Indeed, a vertex first covered at position j contributes $s_{\pi(j)} = u_j + \cdots + u_d$. Integrating the u_k separately gives the product in (3). Permutations here order labeled positions, even when some e_i coincide; all $d!$ chambers must still be included. □

Proof of Theorem 1. All required moments are finite: $0 < Y_e \le \sum_{v \in V} T_v$ almost surely, and a finite sum of exponentials has all integer moments. Expand $(\sum_e x_e Y_e)^d$ as a sum over ordered edge tuples and use Lemma 1. For each fixed $\pi \in S_d$, the relabeling $(e_1, \dots, e_d) \mapsto (e_{\pi(1)}, \dots, e_{\pi(d)})$ is a bijection of E^d and leaves the monomial unchanged. Thus every one of the $d!$ summands equals $f_d(x)$, proving (2). This is a polynomial identity, including signed x.

Differentiating this finite polynomial identity gives

$$\nabla^2 f_d(x) = \frac{1}{(d-2)!} \mathbb{E}\big[(x \cdot Y)^{d-2} Y Y^{\mathsf{T}}\big]. \tag{4}$$

For $d = 2$ the power is identically one. For $x \ge 0$ the weight is nonnegative, so the right side is positive semidefinite. This already proves the conjectured convexity, including the boundary, by restriction to line segments.

For strictness we prove that the functions $t \mapsto \min_{v \in e} t_v$, $e \in E$, are linearly independent on $(0, \infty)^V$. If a linear combination with coefficients a_e vanishes there, continuity extends the identity to $[0, \infty)^V$. At the indicator vector $t = \mathbf{1}_S$, for $S \subseteq V$, it reads

$$\sum_{e \in E:\ e \subseteq S} a_e = 0.$$

Extend a_e by zero to subsets outside E. Induction on $|S|$ (equivalently, Boolean-lattice Möbius inversion) shows every $a_e = 0$. Since the exponential joint density is positive throughout $(0, \infty)^V$, continuity implies that for every nonzero a, $\mathbb{P}(a \cdot Y \neq 0) > 0$. For nonzero $x \ge 0$, we have $x \cdot Y > 0$ almost surely. Equation (4) is therefore positive definite. Alternatively, strict convexity on the entire orthant, including segments from the origin, follows directly from the strict convexity of $t \mapsto t^d$ on $[0, \infty)$: for distinct $x, z \ge 0$, $x \cdot Y \neq z \cdot Y$ with positive probability.

Finally take $E = \binom{[n]}{L}$. Permuting vertices preserves (1) and acts transitively on the edge set. Averaging any $x \in \Delta_m$ over all vertex permutations gives $x^* = (1/m, \dots, 1/m)$. Convexity and invariance imply $f_d(x^*) \le f_d(x)$, and strict convexity gives uniqueness. This last symmetry argument is the one already used in [1]. □

The coefficient matrices in the source. Write $f_d(x) = \sum_{|\alpha| = d} b_\alpha x^\alpha$ and $\alpha! = \prod_e \alpha_e!$. Formula (2) implies $b_\alpha = \mathbb{E}[Y^\alpha]/\alpha!$. Thus the matrices in [1, Lemmas 4.1 and 4.3] satisfy

$$\nabla^2 f_d(x) = \sum_{|\gamma| = d-2} x^\gamma Q_\gamma, \qquad Q_\gamma = \frac{1}{\gamma!} \mathbb{E}[Y^\gamma Y Y^{\mathsf{T}}]. \tag{5}$$

They are positive definite, since $Y^\gamma > 0$ almost surely and the independence argument above applies. This proves the stronger coefficientwise positivity question posed in the source as well. For a single edge of size L, (2) gives $f_d(x) = x^d / L^d$, confirming the factor $d!$.

Independent finite verification. The supplied standard-library Python verifier uses exact rational arithmetic. One algorithm evaluates the original ordered-edge sum by merging prefixes with the same union. A second integrates over the $|V|!$ chambers of ordered *vertex* lifetimes. To describe

the second algorithm, put $N = |V|$, let $T_{\sigma(1)} \le \cdots \le T_{\sigma(N)}$, and define $S_j = \{\sigma(j), \ldots, \sigma(N)\}$, $h_j = \sum_{e \subseteq S_j} x_e$, $r_j = N - j + 1$. Then

$$f_d(x) = \frac{1}{N!} \sum_{\sigma \in S_N} [z^d] \prod_{j=1}^{N} \frac{1}{1 - z h_j / r_j}.$$

Indeed, on this chamber, successive spacings $u_j \ge 0$ have density $\exp(-\sum_j r_j u_j)$ and $x \cdot Y = \sum_j h_j u_j$. Expanding its dth power and integrating uses $\int_0^\infty u^k e^{-ru}\, du = k!/r^{k+1}$; the product of the r_j is $N!$.

The two algorithms agree in 585 exact value comparisons through degree 12 and in 596 symbolic coefficients, including repeated edges, mixed edge sizes, signed inputs, and simplex vertices. All 84 sampled Hessian coefficient matrices have exact positive elimination pivots. Separate checks verify the single-edge normalization and the Boolean inversion used for strictness. These finite checks supplement the unrestricted proof above.

Scope and attribution. The source proves convexity for $d = 2$ and for $(d, L) = (3, 2)$; Polak [2] proves $L = 2$, $d \le 9$. The present contribution is the moment representation and its application for arbitrary d and nonempty edge families. Tail integration, exponential survival probabilities, permutation chambers, and Gram positivity are classical tools. Both the complete current target preprint and its journal final were read. The bounded prior-claim audit and its limits are recorded separately. Review is by the authoring agent only. The minimum corollary and matrix positivity are included in this one result; they receive no separate campaign credit.

Task 2: Find useful explicit lower bounds for the smallest eigenvalue of Q_γ in terms of the hypergraph and γ.

Task 3: Determine sharp quantitative gaps between $f_d(x)$ and its uniform minimum for complete uniform hypergraphs.

Task 4: Use the moment representation to study generating functions in d and their relation to queue stability and occupancy.

Task 5: For unequal vertex service rates, characterize the minimizing edge distribution and the symmetries that force its coordinates.

References

[1] D. Brosch, M. Laurent and A. Steenkamp, Optimizing hypergraph-based polynomials modeling job-occupancy in queuing with redundancy scheduling, *SIAM Journal on Optimization* **31**(3) (2021), 2227–2254. doi:10.1137/20M1369592. Conjecture 1, journal p. 2242; source matrices on pp. 2243–2244. arXiv:2009.04510v1, Conjecture 1 on preprint p. 20.

[2] S. C. Polak, Symmetry reduction to optimize a graph-based polynomial from queueing theory, *SIAM Journal on Applied Algebra and Geometry* **6** (2022), 243–266. doi:10.1137/21M1413298. arXiv:2104.08264v2, Theorem 1.1.